

Bird's-Eye View Semantic Segmentation for Autonomous Driving via Limited Multi-Modal Sensor Fusion

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Abstract— This paper investigates whether a single front camera and a single LiDAR are sufficient for Bird's-Eye View (BEV) semantic segmentation on cost-constrained autonomous platforms. State-of-the-art BEV methods rely on six surround cameras and dense LiDAR, which is impractical for cost-constrained vehicles such as MEVi, a Micro Electric Autonomous Vehicle. We propose a multi-modal fusion network combining an EfficientNetV2-S camera backbone with a SECOND-based LiDAR backbone, integrated through a three-stage cross-attention mechanism. Trained on a subset of the nuScenes dataset and evaluated on the complete official nuScenes validation split, the model achieves 49.4% mean Intersection over Union (mIoU), exceeding the six-camera LSS baseline (44.4%) and approaching the six-camera-plus-LiDAR BEVFusion method (56.6%). Drivable area is detected reliably (83.7% IoU); thin structures such as lane and road dividers remain challenging (23.6% and 34.3% IoU). These results demonstrate that minimal-modality BEV perception is viable for navigable space detection on resource-constrained autonomous vehicles, with a model whose fusion pipeline accounts for less than 4% of total trainable parameters.

Keywords— Autonomous Driving, Bird's-Eye View (BEV), Semantic Segmentation, Sensor Fusion, Multi-Modal Fusion, Cross-Attention, Deep Learning.

I. INTRODUCTION

Autonomous driving relies on accurate spatial perception to navigate safely. Bird's-Eye View (BEV) representation offers a top-down perspective of the environment that directly supports motion planning, lane-following, and obstacle avoidance [1]. Generating dense BEV semantic maps is therefore a critical task for any autonomous platform.

Current state-of-the-art BEV methods achieve high performance by combining comprehensive sensor suites: typically six or more surround cameras plus a high-density LiDAR. Lift-Splat-Shoot (LSS) [2] implicitly lifts perspective-view camera features into 3D and achieves 44.4% mIoU using six cameras. BEVFusion [3] fuses six-camera and LiDAR BEV features through a unified representation, reaching 56.6% mIoU. BEVFormer [4] employs spatial-temporal transformers over six camera views. These approaches demonstrate the value of sensor redundancy, but their hardware requirements make them unsuitable for compact, cost-constrained platforms.

This study addresses a gap that has received little attention: BEV segmentation with fewer than six cameras. Specifically, we target MEVi, a Micro Electric Autonomous Vehicle equipped with only a single front-facing camera and a single top-mounted LiDAR. The MEVi platform represents a class of low-cost autonomous vehicles where sensor reduction is not a choice but a hard constraint. To our knowledge, no published work directly evaluates BEV segmentation in this single-camera, single-LiDAR setting on the nuScenes benchmark.

We propose a multi-modal fusion network tailored to this constrained setup. The camera branch uses EfficientNetV2-S [5] pre-trained on ImageNet [6] to extract multi-scale features. The LiDAR branch uses a SECOND-based backbone [7] with sparse convolutions for efficient BEV feature extraction. Both branches are fused through a three-stage cross-attention mechanism in which image features query LiDAR features, grounding semantic richness with spatial precision. The fused features are

passed to a lightweight segmentation head trained with focal and dice losses to handle severe class imbalance.

The main contributions of this work are: (1) the first evaluation of the 1-camera + 1-LiDAR sensor configuration for BEV semantic segmentation on nuScenes; (2) a lightweight cross-attention fusion architecture whose fusion pipeline adds less than 4% to total parameter count; and (3) an analysis of the performance trade-offs of sensor-limited BEV perception for motion planning on compact autonomous vehicles.

II. RELATED WORKS

A. Camera-Only BEV Methods

LSS [2] defines a pillar-based unprojection that implicitly models depth to lift perspective features into a 3D frustum, then splats them onto the BEV plane. SimpleBEV [8] shows that a simple bilinear sampling with explicit camera geometry achieves competitive results without complex depth networks. BEVFormer [4] uses deformable attention over spatial-temporal camera features, achieving state-of-the-art camera-only performance. All of these methods require at least six surround cameras to maintain full 360-degree coverage.

B. Camera-LiDAR Fusion BEV Methods

BEVFusion [3] projects camera features into BEV space using predicted depth and fuses them with voxelized LiDAR BEV features through channel concatenation. DeepFusion [9] uses an iterative attention mechanism to refine LiDAR features with camera context. These methods demonstrate the complementarity of the two modalities but still rely on multi-camera setups for spatial coverage.

C. Constrained Sensor Setups

Most BEV segmentation benchmarks evaluate on the full nuScenes sensor suite. Research on reduced-sensor configurations is sparse. Some work has explored camera-only setups with a reduced number of views [8], to our knowledge, no published study directly targets the 1-camera + 1-LiDAR configuration for BEV semantic segmentation. This work fills that gap, providing a baseline for perception on compact autonomous platforms.

III. METHODS

The proposed pipeline has four stages: (A) data preprocessing, (B) camera feature extraction, (C) LiDAR feature extraction, (D) cross-attention fusion, (E) segmentation head and loss, (F) training configuration, and (G) evaluation protocol.

A. Data Preprocessing

We use the nuScenes dataset [10], filtering samples to retain only those with both a front camera (CAM_FRONT) and top LiDAR (LIDAR_TOP). BEV segmentation labels are generated offline by rasterizing HD map data from the nuScenes Map API into a 128×128 pixel grid at 0.8 m/pixel resolution, covering a 102.4 m × 102.4 m area centered on the ego vehicle. The ego vehicle is placed at the bottom center of the grid to form a rear-cropped BEV map, ensuring the front field of view aligns with the single camera's coverage. Six semantic classes are defined:

Background, Drivable Area, Lane Divider, Road Divider, Walkway, and Pedestrian Crossing. Fig. 1 shows representative BEV label examples.

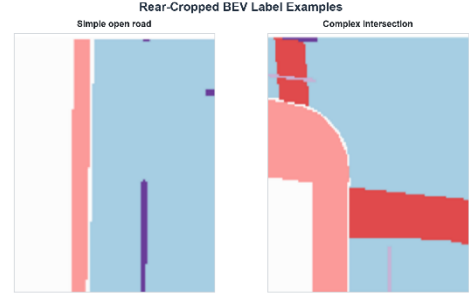


Fig. 1. Rear-cropped BEV label examples from the nuScenes dataset. Left: a simple open road. Right: a complex intersection with multiple road elements.

B. Feature Extraction

For the camera branch, EfficientNetV2-S [5] pre-trained on ImageNet [6] is used as the backbone. The network is modified using torchvision's feature extractor API to produce multi-scale feature maps at five stages. Projection layers (Conv2d 1×1, BatchNorm, ReLU) are applied at each stage to standardize channel dimensions to 32, 64, 128, 256, and 512 for stages 1 through 5, respectively. This reduces memory usage and ensures compatibility with the fusion module. Stages 3, 4, and 5 (channels 128, 256, and 512) are passed to the fusion module.

For the LiDAR branch, a SECOND-based backbone [7] with sparse convolutions (spconv) [11] is used. Raw LiDAR point clouds are voxelized into a 0.8 m × 0.8 m × 0.8 m grid within the range [-51.2, 51.2] m in X and Y, and [-5, 3] m in Z. A multi-branch linear voxel encoder computes 10-dimensional per-voxel features (mean xyz, intensity, geometric statistics) and projects them to 32 channels. Three sparse encoder stages progressively expand the channels to 64, 128, and 256. After each stage, a BEV projection converts the sparse 3D tensor to a dense 2D feature map by max-pooling along the Z axis and applying a lightweight depthwise convolution block. This produces BEV feature maps at three scales (64, 128, 256 channels), denoted Stage 1, Stage 2, and Stage 3.

C. Multi Modal Fusion

To integrate the two modalities, a three-stage cross-attention fusion module is proposed. At each stage $s \in \{1, 2, 3\}$, the image features I_s are projected to a common channel dimension $C_{out} = 128$ and serve as queries Q , while the LiDAR BEV features L_s are projected and serve as keys K and values V . Spatial dimensions are aligned by bilinear interpolation. The attention output is computed as:

$$\text{Fused}_s = \text{softmax}\left(\frac{QK^T}{\sqrt{C}}\right)V + \lambda^0 I_s$$

where C is the channel dimension and λ^0 is a learnable scalar weight. To reduce memory, the spatial attention is computed over chunks of 2048 pixels with gradient checkpointing. The fused outputs from all three stages are concatenated and passed through a feature aggregation network (two convolutional layers, upsampled to the target

128×128 BEV resolution) to produce a 128-channel unified BEV feature map. Fig. 2 illustrates the complete architecture.

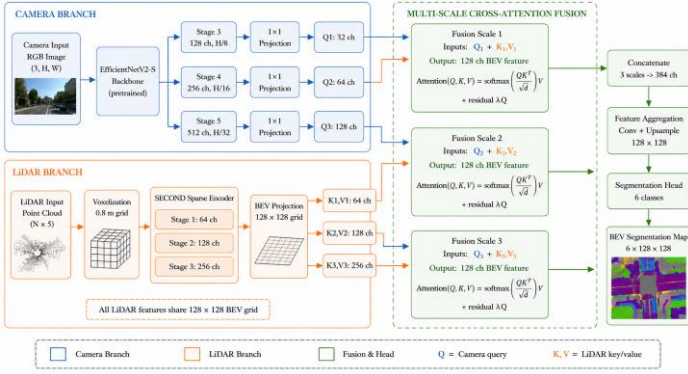


Fig. 2. Proposed model architecture. The camera branch (blue) extracts multi-scale features via EfficientNetV2-S. The LiDAR branch (orange) produces BEV features via a SECOND-based sparse encoder. Both branches are fused through three cross-attention stages (green) where image features query LiDAR features. The segmentation head predicts a 128×128 six-class BEV map.

D. Segmentation Head and Loss

The segmentation head is a lightweight two-layer network: a 3×3 grouped convolution (8 groups), batch normalization, ReLU, dropout (rate 0.2), followed by a 1×1 convolution that maps 128 channels to 6 class logits. During training, the head receives a 128-channel fused BEV map and produces per-pixel class logits.

Two losses are combined in equal proportion. Focal Loss [12] focuses training on hard pixels and down-weights easy negatives, addressing class imbalance with $\gamma = 2.0$. Dice Loss [13] directly optimizes the IoU metric for unbalanced classes. Static per-class weights [0.75, 0.5, 5.0, 5.0, 2.0, 1.0] further up-weight minority classes. Label smoothing ($\epsilon = 0.05$) is applied during training. The total loss is:

$$\mathcal{L}_{total} = 0.5 \times \mathcal{L}_{focal} + 0.5 \times \mathcal{L}_{dice}$$

E. Training Configuration

The model is trained end-to-end with the AdamW optimizer using four separate learning-rate groups for the camera backbone, LiDAR backbone, fusion module, and segmentation head (all at 5×10^{-5} , weight decay 1×10^{-2}). A Cosine Annealing schedule decays the rate to 1×10^{-6} over 54 epochs. Gradient clipping is applied at a maximum norm of 1.0. Mixed-precision training (FP16) with GradScaler reduces memory usage. The batch size is 4, with 8 data-loader workers.

Regularization includes: dropout (0.2) in the segmentation head, weight decay (1×10^{-2}), label smoothing (0.05), and BEV data augmentation (random horizontal/vertical flips, rotation $\pm 10^\circ$, scaling, translation, cutout, mixup, and Gaussian noise). An early stopping callback monitors validation mIoU with patience 5, saving the best checkpoint automatically. All experiments use a single NVIDIA GeForce RTX 3080 GPU (10 GB VRAM).

F. Evaluation

We follow the official nuScenes train/val split. Due to local storage and training-time constraints, the training set used in this study is a contiguous 50% subset of the

nuScenes training split (~14,000 samples from 350 scenes). The validation set is the complete official nuScenes validation split (6,019 samples, 150 scenes), identical to the split used by LSS [2] and BEVFusion [3]. No separate test set is used; reported numbers are validation-set results. This means the comparison in Table I differs only in the training data size (half vs. full); the evaluation set is identical and the comparison is therefore valid. The primary metric is mean Intersection over Union (mIoU), averaged over all six classes:

$$IoU_c = \frac{TP_c}{(TP_c + FP_c + FN_c)}$$

$$mIoU = \left(\frac{1}{Class} \right) \sum_{class} IoU_{class}$$

Per-class precision and recall are additionally reported via the normalized pixel-wise confusion matrix.

IV. RESULTS AND DISCUSSION

A. Learning Behaviour

The model was trained for 138 epochs. Fig. 3 shows the training and validation loss curves. Both losses decrease sharply in the first 20 epochs, indicating rapid learning of fundamental BEV features. From epoch 20 onward, training loss continues to decrease while validation loss plateaus near 0.94, signaling the onset of overfitting.



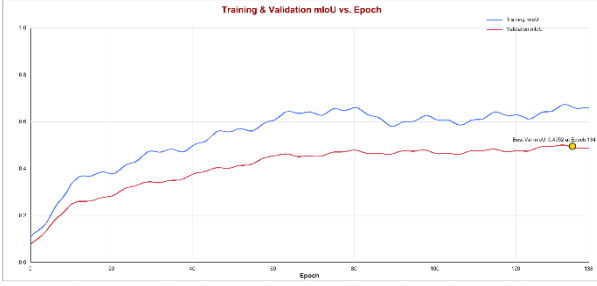
Fig. 3. Training and validation loss curves over 138 epochs. Validation loss plateaus near epoch 70, indicating the onset of overfitting.

B. Generalization Gap

Fig. 4 shows the training and validation mIoU curves. The training mIoU reaches 66.2% at epoch 138. The validation mIoU plateaus earlier, peaking at 49.5% (0.4952) at epoch 134. This 16-point gap is a clear sign of overfitting. The gap is attributable to three factors: (1) the training set is half the full nuScenes training split, limiting data diversity; (2) EfficientNetV2-S has high capacity relative to the available training data; and (3) thin classes such as lane and road dividers appear sparsely, making their statistics harder to generalize.

Several regularization strategies are already applied: dropout (0.2), weight decay (1×10^{-2}), label smoothing (0.05), and BEV augmentation. Early stopping with patience 5 selects the model at epoch 134 as the optimal deployment checkpoint. Despite these measures, the gap persists, and training on the full nuScenes training split is the most direct path to improvement.

Fig. 4. Training and validation mIoU curves. The best validation checkpoint is at epoch 134 (mIoU = 49.5%), selected by early stopping.



The 16-point generalization gap is primarily attributed to training on a 50% data subset.

C. Validation Performance

Table I reports class-wise and mean IoU at the best validation checkpoint (epoch 134). The proposed 1C+1L model achieves 49.4% mIoU on the full official nuScenes validation set, exceeding the six-camera LSS baseline (44.4%) and approaching the six-camera-plus-LiDAR BEVFusion (56.6%). These comparisons should be interpreted with care: our model uses a smaller training set and a more constrained sensor configuration. The table is contextual, not a claim of outperformance, and the footnote marks the training-data difference explicitly.

The model excels at large, dominant classes. Drivable Area reaches 83.7% IoU on the validation set, indicating robust detection of navigable space, which is the most critical output for motion planning on MEVi. Background is identified at 62.9% IoU. Performance drops sharply for thin, geometrically complex structures: Lane Divider achieves 23.6% IoU and Road Divider 34.3% IoU. These classes are narrow (often 1-2 pixels wide in the BEV map) and sparsely distributed.

Qualitative results are shown in Figs. 5 and 6. Training-set predictions (Fig. 5) demonstrate near-perfect pixel-level replication of ground truth, confirming the model’s high memorization capacity. On the validation set (Fig. 6), drivable area is consistently detected across scenes. In the simple open-road case (Fig. 6a), road dividers are largely correct, but the thinner lane divider shows fragmentation. In the complex traffic scene (Fig. 6b), the pedestrian crossing appears as a scattered cluster of pixels, and the lane divider is partially segmented. These predictions are free from the diagonal streaking artifacts observed in alternative image-to-BEV projection architectures, confirming the stability of the heuristic alignment approach for this constrained sensor configuration.

TABLE I. CLASS-WISE AND MEAN INTERSECTION OVER UNION (IOU) COMPARISON

Methods	Modality	Training data	Background	Drivable Area	Lane Divider	Road Divider	Walkway	Ped.Cross	Mean
LSS [2]	6C	Full	-	75.4	19.96	-	46.3	38.8	44.4
BEVFusion [3]	6C+1L	Full	-	85.5	46.4	-	67.6	60.5	56.6
Proposed (train)	1C+1L	50% subset*	75.3	88.2	35.9	44.2	72.2	70.1	64.3
Proposed (val)	1C+1L	50% subset*	62.9	83.7	23.6	34.3	40.9	51.4	49.4

*Trained on ~50% of nuScenes train split (~14k samples). All results evaluated on the full official nuScenes val split (6,019 samples, 150 scenes).

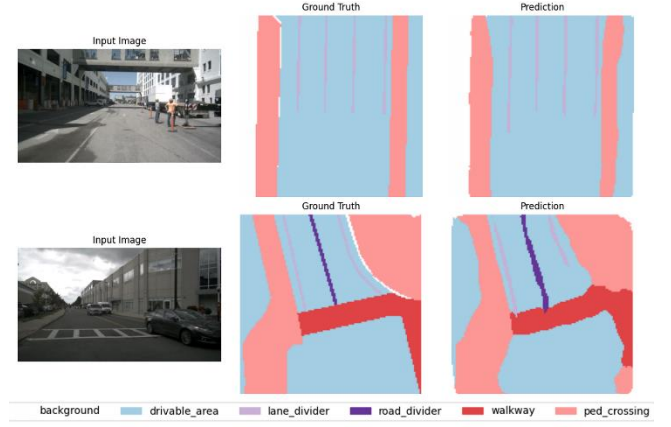


Fig. 5. Segmentation predictions on the training set. (a) Simple open-road scenario. (b) Complex scene with pedestrian crossing and road/lane dividers. The model shows high memorization capacity on training data.

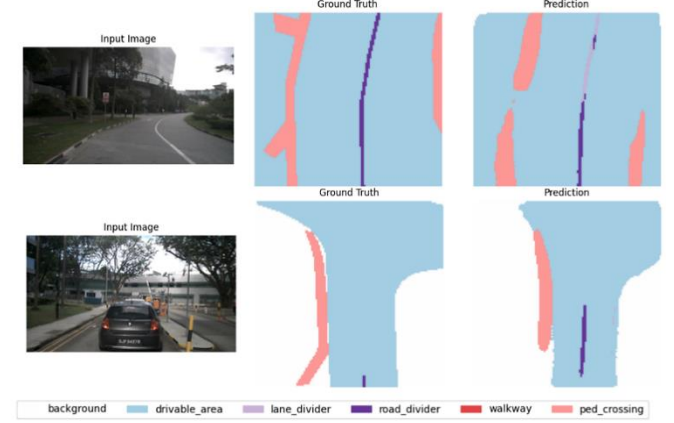


Fig. 6. Segmentation predictions on the validation set. (a) Clear open-road sample (Fig. 6a). (b) Moderate traffic with pedestrian infrastructure (Fig. 6b). Drivable area is consistently detected; fine structures show degradation.

D. Confusion Matrix Analysis

The normalized pixel-wise confusion matrix at the best checkpoint (epoch 134) is shown in Fig. 7. Rows represent true classes; columns represent predicted classes; the diagonal gives per-class recall.

Drivable Area has the highest recall (90.3%), confirming robust navigable-space detection. Background recall is 81.0%. Classes related to pedestrian infrastructure show moderate recall: Walkway 57.3% and Pedestrian Crossing 67.1%, with frequent misclassification as Drivable Area.

This suggests the model relies on local texture rather than scene-level context to separate walkable from drivable surfaces. The weakest classes are Road Divider (51.0% recall) and Lane Divider (36.9% recall), both frequently confused with Drivable Area. This failure mode is consistent with the geometric challenge of thin, high-aspect-ratio structures at 0.8 m resolution.

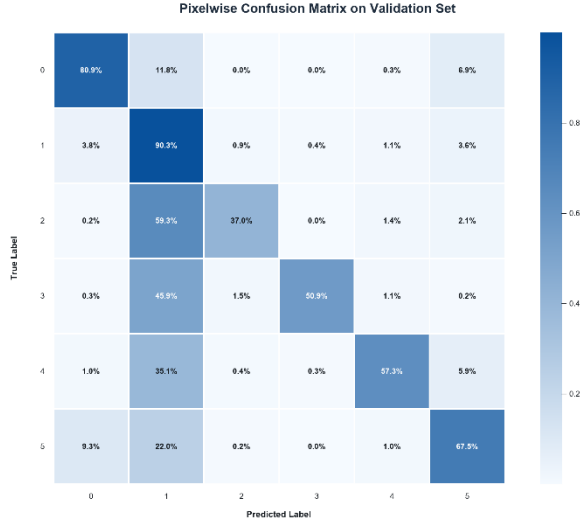


Fig. 7. Normalized pixel-wise confusion matrix on the validation set at the best checkpoint (epoch 134). Rows: true class. Columns: predicted class. Diagonal: per-class recall.

E. Computation

For compact autonomous platforms like MEVi, inference speed and memory footprint are as important as accuracy. All experiments ran on a single NVIDIA RTX 3080 GPU (10 GB VRAM) with batch size 4. A full training epoch over the 14,000-sample training subset took approximately 2.5 hours (~9,100 seconds). Validation over the 6,019-sample set completed in approximately 11 minutes (~660 seconds).

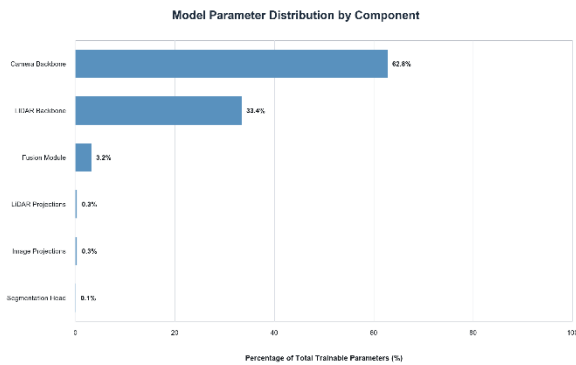


Fig. 8. Model parameter distribution by component. The proposed fusion pipeline (cross-attention, projections, segmentation head) accounts for less than 4% of total trainable parameters

Fig. 8 shows the parameter distribution across model components. The camera and LiDAR backbones account for 62.8% and 33.4% of total parameters, respectively. The entire fusion pipeline—including the three-stage cross-attention module, projection layers, and the segmentation head—contributes less than 4% of total parameters. This

confirms that the proposed fusion approach is computationally lightweight and that model size is determined primarily by the backbone choice. Replacing EfficientNetV2-S with a smaller EfficientNet variant would reduce total parameters with modest accuracy trade-off.

V. CONCLUSION

This study shows that BEV semantic segmentation is achievable with only a single front camera and a single LiDAR, a sensor configuration common on cost-constrained autonomous platforms such as MEVi. The proposed cross-attention fusion network achieves 49.4% mIoU on the full official nuScenes validation set, exceeding the six-camera LSS baseline (44.4%) while using substantially fewer sensors. Drivable area detection, the most critical output for motion planning, reaches 83.7% IoU. The entire fusion pipeline adds less than 4% to total model parameters.

The main limitations are: (1) a 16-point generalization gap between training and validation mIoU, driven by the 50% training data subset and the inherent difficulty of thin road elements; and (2) weak performance on Lane and Road Dividers (23.6% and 34.3% IoU). Future work should train on the full nuScenes training set, explore lighter backbone alternatives for embedded deployment, and investigate temporal fusion to improve thin-structure recall. Integrating explicit camera-to-BEV depth estimation (e.g., via LSS-style lifting) could further improve spatial alignment between modalities

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